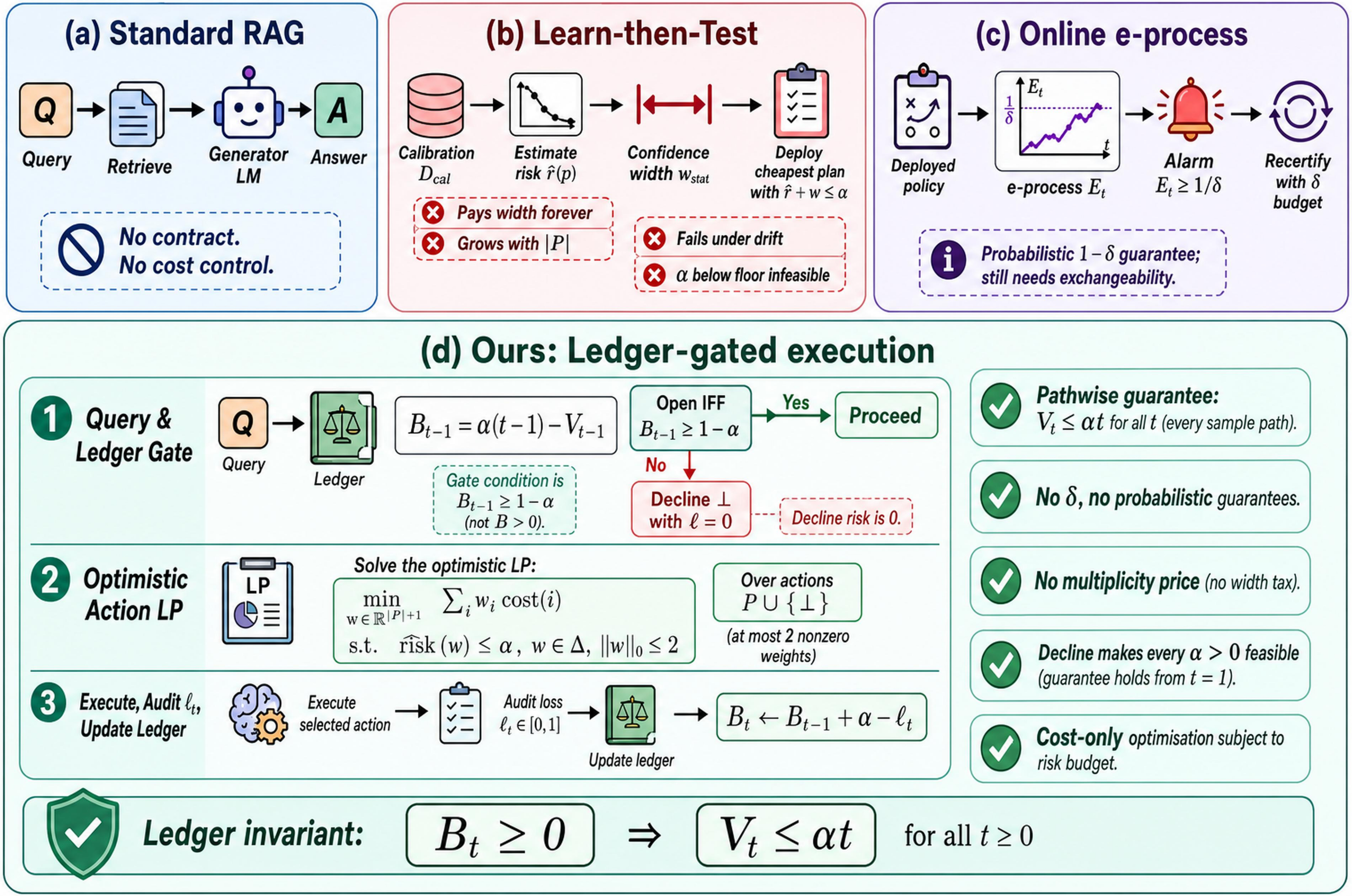
(a) Standard RAG
Q
Query
Retrieve
Generator LM
A
Answer
No contract.
No cost control.
(b) Learn-then-Test
Calibration $D_{cal}$
Estimate risk $\hat{r}(p)$
Confidence width $w_{stat}$
Deploy cheapest plan with $\hat{r} + w \le \alpha$
Pays width forever
Grows with $|P|$
Fails under drift
$\alpha$ below floor infeasible
(c) Online e-process
Deployed policy
$E_t$
$\frac{1}{\delta}$
$t$
e-process $E_t$
Alarm $E_t \ge 1/\delta$
Recertify with $\delta$ budget
Probabilistic $1-\delta$ guarantee; still needs exchangeability.
(d) Ours: Ledger-gated execution
1 Query & Ledger Gate
Q
Query
Ledger
$B_{t-1} = \alpha(t-1) - V_{t-1}$
Open IFF $B_{t-1} \ge 1-\alpha$
Yes
Proceed
No
Decline $\perp$ with $\ell = 0$
Decline risk is 0.
Gate condition is $B_{t-1} \ge 1-\alpha$ (not $B > 0$).
2 Optimistic Action LP
LP
Solve the optimistic LP:
$\min_{w \in \mathbb{R}^{|P|+1}} \sum_i w_i \operatorname{cost}(i)$
s.t. $\widehat{\text{risk}}(w) \le \alpha,\ w \in \Delta,\ \|w\|_0 \le 2$
Over actions $P \cup \{\perp\}$
(at most 2 nonzero weights)
3 Execute, Audit $\ell_t$, Update Ledger
Execute selected action
Audit loss $\ell_t \in [0,1]$
Update ledger
$B_t \leftarrow B_{t-1} + \alpha - \ell_t$
Pathwise guarantee: $V_t \le \alpha t$ for all $t$ (every sample path).
No $\delta$, no probabilistic guarantees.
No multiplicity price (no width tax).
Decline makes every $\alpha > 0$ feasible (guarantee holds from $t = 1$).
Cost-only optimisation subject to risk budget.
Ledger invariant: $B_t \ge 0 \Rightarrow V_t \le \alpha t$ for all $t \ge 0$